AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

MIS200-IS

Relational Database Management Systems

Final Examination / Version: B

2025-2026 / SPRING

8 – June – 2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 hours**

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF"** mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
8- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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YOU HAVE TO ANSWER ALL THE QUESTIONS

Question 1 (35 marks)

Construct an E-R diagram for a hotel booking system that includes rooms and guests. Associate with each guest a history of their reservations, including room details, reservation dates (check-in/check-out), and payment status. After drawing the E-R diagram, develop the relational schema (tables) and clearly indicate the PKs and FKs used to implement the relationships.

Question 2 (35 marks)

Consider the following clinic appointment database, where the primary keys are underlined.

customer (customer_id, name, phone)
vehicle (vehicle_id, plate_number, model)
rental (rental_id, customer_id, vehicle_id, rental_date, return_date, payment_status)

a) Write DDL statements for creating all tables with constraints. Make any reasonable assumptions about data types and be sure to declare primary and foreign keys (15 marks).

b) Write an SQL query to retrieve rentals for a customer named 'Sara Ali' (5 marks).

c) Write an SQL query to insert a new customer record (assume the values) (5 marks).

d) Write an SQL query to delete a rental using rental_id (assume the rental_id is known) (5 marks).

e) Write an SQL query to remove the entire vehicle table from the database (5 marks).

Question 3 (30 marks)

a) Transform the following table into Second Normal Form (2NF) (20 marks):

PatientID	PatientName	DoctorID	DoctorSpecialization	VisitDate
1	Sara	10	Cardiology	2024-06-10
2	Omar	11	Dermatology	2024-06-12

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b) Choose the correct answer for each question below (10 marks):

1) Given the table

CUSTOMER(CustID, Name, PhoneNum, AcctBalance)

What is the standard SQL query phrase to retrieve the Name and Phone Number of customers?

A) SELECT CUSTOMER-Name AND CUSTOMER-PhoneNum
B) SELECT (CUSTOMER-Name AND CUSTOMER-PhoneNum)
C) SELECT Name, PhoneNum
D) SELECT (Name, PhoneNum)

2) Which of the following data types used in SQL would define a fixed-length text field of 10 characters?

A) Text(10)
B) Char(10)
C) VarDhar(10)
D) Fixed(10)

3) The number of entity classes involved in a relationship is known as the relationship's _____.

A) maximum cardinality
B) minimum cardinality
C) ordinality
D) degree

4) Which of the following is not a stage in the development of a database system?

A) Requirements analysis
B) Validation
C) Component design
D) Implementation

5) In many-to-many relationships in a relational database design, _____.

A) the key of the child is placed as a foreign key into the parent
B) the key of the parent is placed as a foreign key into the child
C) the keys of both tables are placed in only one of the tables
D) the keys of both tables are joined into a composite key

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